

Grid Resolution Closure

Motivation and Challenges

Motivation:

Fully resolved physics:

- High accuracy at High Cost
- Need for reduction

Under-resolved physics:

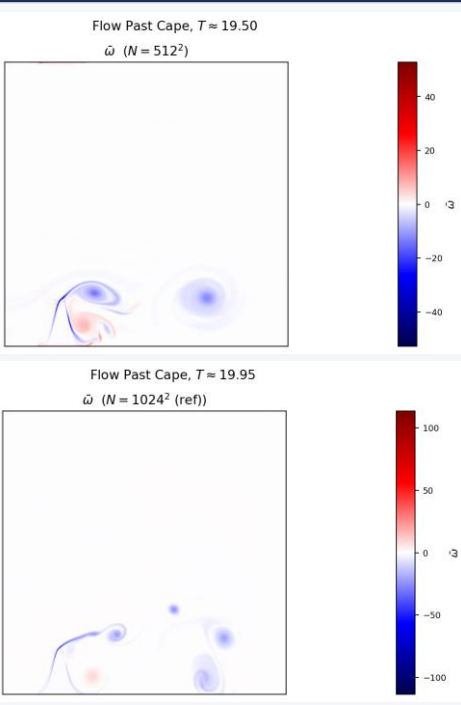
- Lower accuracy at Lower Cost
- Truncation / discretization error

Challenges:

- Complex nonlinear, multiscale fluid dynamics with unresolved sub-grid physics
- Truncated numerical models introduce truncation and discretization biases
- Regime variability (Reynolds-number shifts, transient forcing) and evolving flow structures

Needs:

- Fast, online data-driven correction of low-resolution dynamics, embedded directly into the solver
- Uncertainty-aware predictions of resolved fields and wake metrics
- Transferable across resolutions, domains, and flow conditions



Flow Past an Idealized Coastal Cape at different resolutions

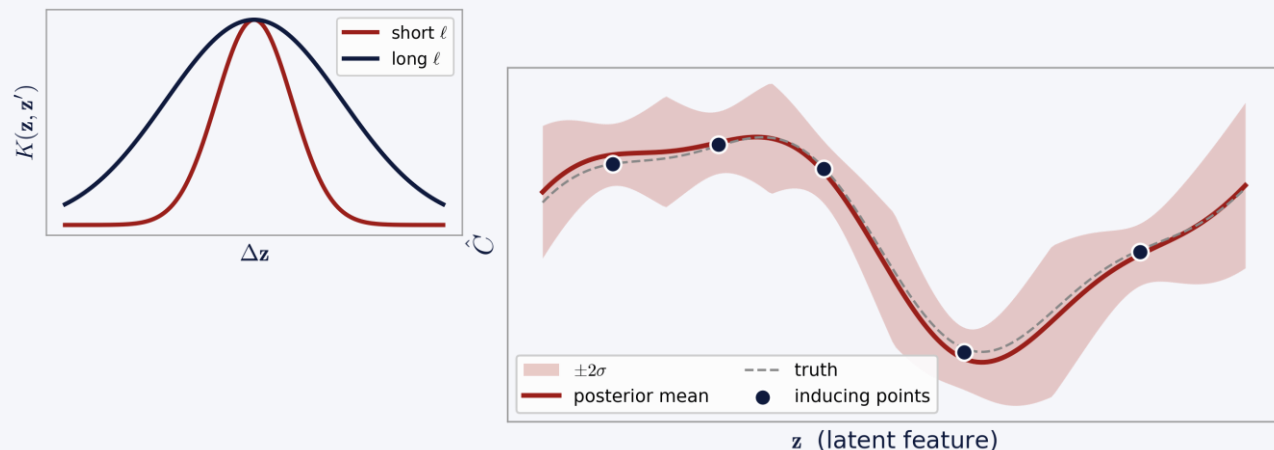
Deep Gaussian Process

SVGP on deep features

$$\mathbf{z}_{i,j} = \text{CNN}(\Phi)_{i,j}$$

$$\hat{C}_{i,j} \sim \mathcal{GP}(\mu(\mathbf{z}_{i,j}), K(\mathbf{z}_{i,j}, \mathbf{z}'_{i,j}))$$

- Structured priors via the kernel (smoothness, length-scales)
- Non-parametric — no rigid analytic form
- Built-in uncertainty: posterior σ^2 per pixel
- Inducing points + ELBO scale to many DOFs



Methodology

(1) Navier-Stokes equations

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0$$

(2) Discrete operators on each grid

Fine grid (h)

$$N_h(\mathbf{u}_h, p_h) = A_h(\mathbf{u}_h) + G_h p_h - \nu L_h \mathbf{u}_h - \mathbf{f}_h$$

high resolution; resolves all scales

Coarse grid (H)

$$N_H(\bar{\mathbf{u}}, \bar{p}) = A_H(\bar{\mathbf{u}}) + G_H \bar{p} - \nu L_H \bar{\mathbf{u}} - \mathbf{f}_H$$

low resolution; cheap but biased

(3) Closure decomposition

$$\hat{\mathbf{C}} = (R A_h \mathbf{u}_h - A_H \bar{\mathbf{u}}) + (R G_h p_h - G_H \bar{p}) - \nu (R L_h \mathbf{u}_h - L_H \bar{\mathbf{u}}) - (R \mathbf{f}_h - \mathbf{f}_H)$$

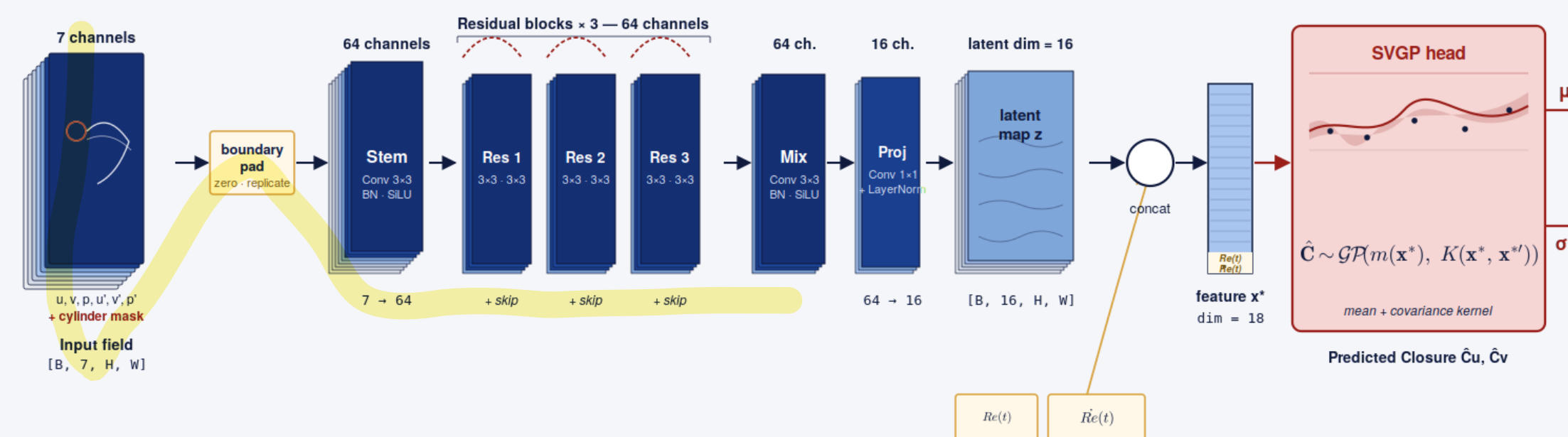
⇒ Learn $\hat{\mathbf{C}} \approx \mathbf{C}$ as a Deep Gaussian Process closure

trained on snapshot pairs $(\mathbf{u}_h, \mathbf{C})$ and added to the coarse RHS at every step

$$R = \text{restriction (fine} \rightarrow \text{coarse)}; \mathbf{u}_H = R \mathbf{u}_h$$

Architecture

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{Re(t)} \nabla^2 \mathbf{u} + \hat{\mathbf{C}}, \quad \nabla \cdot \mathbf{u} = 0$$

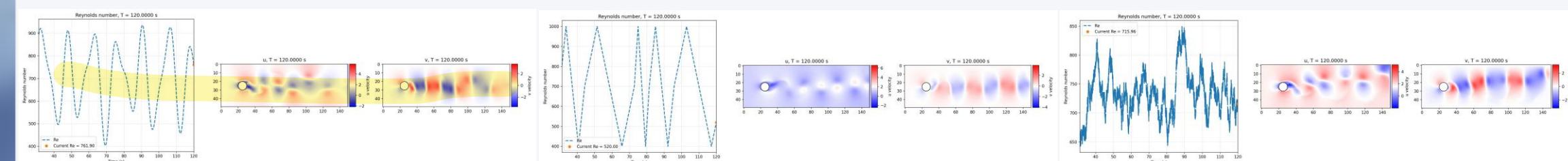


Training: Re(t) Modulation Families

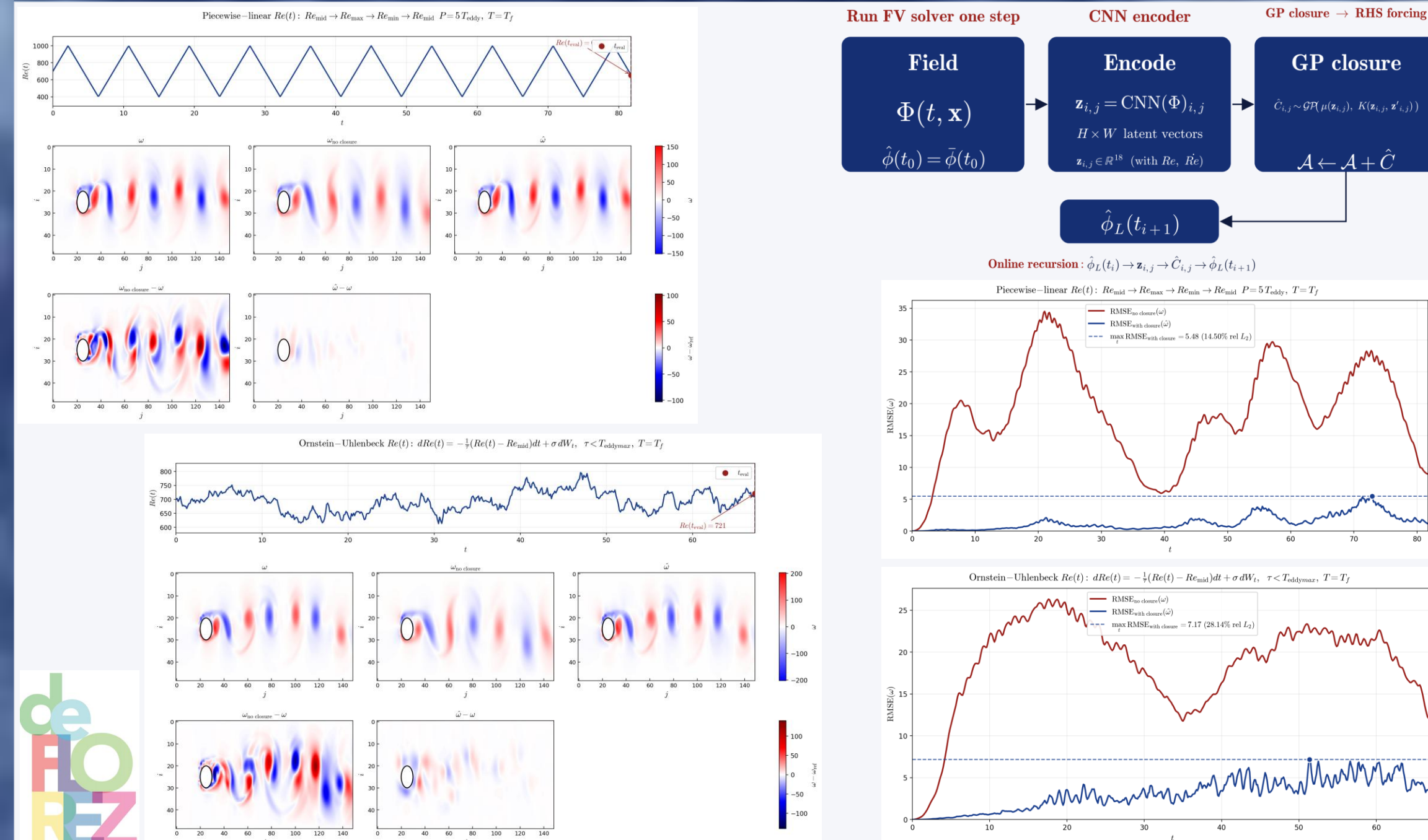
Deterministic broadband (multi-sine Re(t))

Deterministic narrowband (multi-Period triangular wave Re(t))

Stochastic (Ornstein-Uhlenbeck Re(t))



Inference Results: Out-Of-Distribution Rollouts



Step-size Resolution Closure

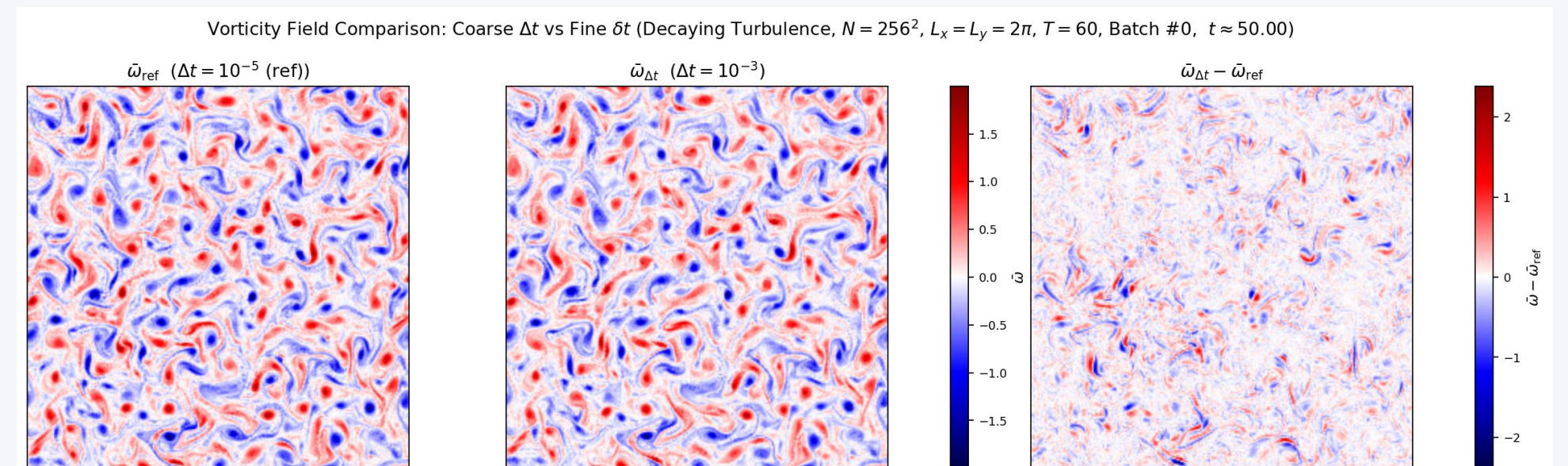
Motivation

Why a temporal closure?

- Spatial closure trains on snapshot pairs — no constraint on temporal coherence
- Coarse ΔT introduces its own truncation error, independent of grid resolution
- Errors at step k feed back into step $k+1$: rollouts can drift over long horizons
- Goal: a closure consistent in time that matches a fine-step reference at coarse cost

Approach:

- Solver of choice : AB2CN2, spectral solver
- Match the local truncation error of K fine AB2CN2 steps ($dt = \Delta T/K$) with one coarse ΔT step
- Closed-form additive RHS correction δR : three terms are pointwise spectral, one (l) is a neural surrogate
- AB2CN2 at ΔT behaves like AB2CN2 at $\Delta T/K$, at K -fold lower runtime cost



Methodology

(1) Truncation error matching — fine reference vs. coarse step

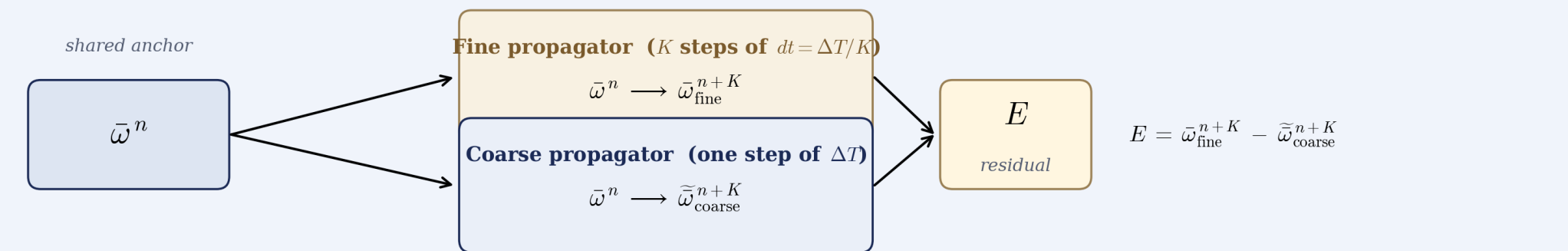
$$\bar{\omega}_{\text{coarse}}^1 - \bar{\omega}_{\text{fine}}^K = \Delta T^3 \left(1 - \frac{1}{K^2}\right) \mathcal{E}(\bar{\omega}^0) + \mathcal{O}(\Delta T^4)$$

$$\text{with } \mathcal{E}(\bar{\omega}^0) = \frac{1}{12} [L^3 \bar{\omega} + L^2 N + L \dot{N} - 5 \ddot{N}]$$

⇒ Match by adding δR to the coarse RHS:

$$\delta R^* = \frac{1 - \frac{\Delta T}{2}}{\Delta T} \cdot \frac{\Delta T^2}{12} \left(1 - \frac{1}{K^2}\right) [L^3 \bar{\omega} + L^2 N + L \dot{N} - 5 \ddot{N}]$$

(2) Training-data construction

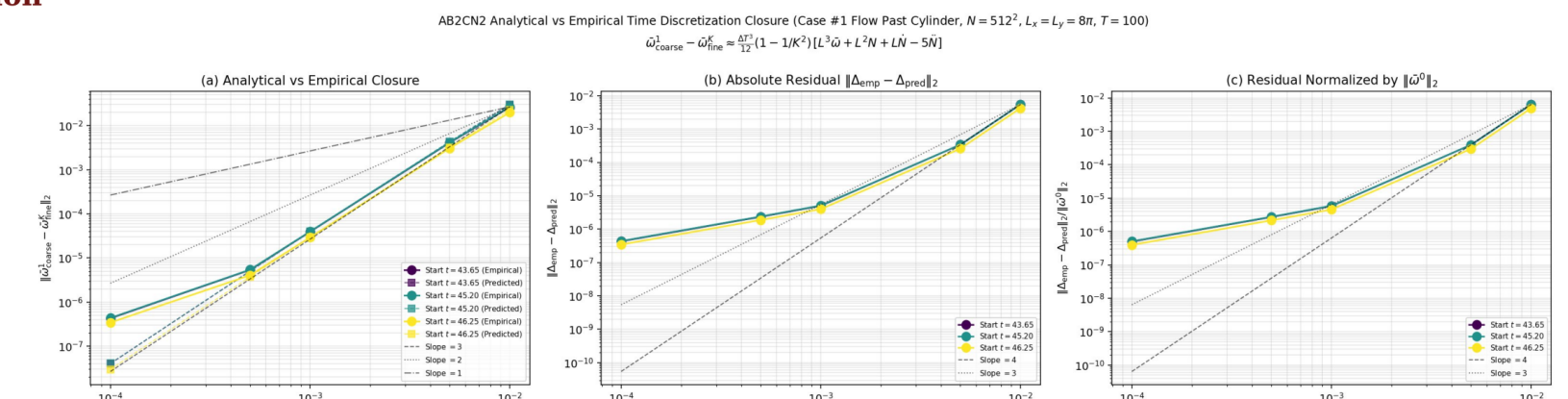


Pointwise-spectral / neural-surrogate decomposition

$$E = E_{\text{lin}}(\bar{\omega}^n) + E_{\text{lin}}(\bar{\omega}^n)$$

with E_{lin} : closed-form pointwise spectral ($L^3 \bar{\omega}, L^2 N, L \dot{N}$) and $E_{\text{lin}} = f_{NN}(\theta)(\bar{\omega}^n, \bar{\psi}^n) \propto \bar{N}$

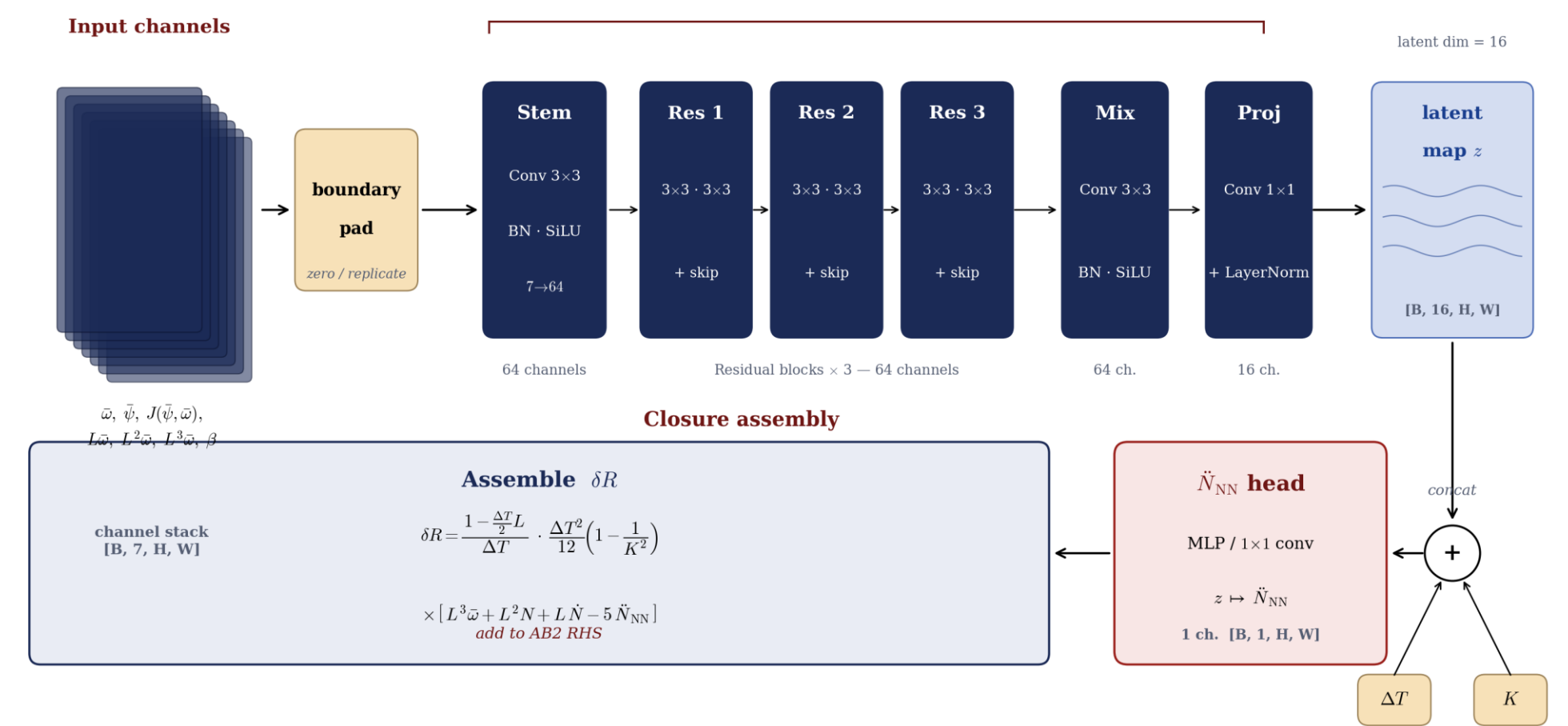
(3) Validation



Architecture

$$\partial_t \bar{\omega} = L \bar{\omega} + N(\bar{\omega}, \bar{\psi}) + \delta R_\theta(\bar{\omega}, \bar{\psi}; \Delta T, K)$$

$f_{NN}(\theta)$: CNN backbone



Online recursion : $\bar{\omega}^n \rightarrow \mathbf{z}_{i,j} \rightarrow \bar{N}_{NN} \rightarrow \delta R \rightarrow \bar{\omega}^{n+1}$

Conclusions and Acknowledgments

Conclusions

- Sparse variational GP closures, trained on snapshot pairs, inject mean (or stochastic) corrections directly into the FV solver and yield stable OOD rollouts on Re(t) modulations the model never saw at training.
- An analytical δR for AB2CN2 IMEX is derived: three pointwise-spectral terms + one NN surrogate (l). Empirical residuals match the predicted closure to high accuracy on flow past a cylinder.
- Together: the spatial closure mitigates grid-resolution bias; the temporal closure mitigates step-size bias. Combined, they target an end-to-end fast, accurate, uncertainty-aware coarse solver.

Acknowledgments

We are grateful to the Office of Naval Research for partial support under grant N00014-20-1-2023 (MURI ML-SCOPE) to MIT. I gratefully acknowledge the MIT de Florez Competition for the opportunity to present this work and for supporting student innovation in mechanical engineering.
Thanks to the MSEAS group and PI Pierre Lermusiaux for their collaboration !